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MCS–203

POST GRADUATE DIPLOMA IN COMPUTER APPLICATIONS (PGDCA) (NEW)

Term-End Examination

December, 2023

MCS-203 : OPERATING SYSTEMS

Time : 3 Hours *Maximum Marks : 100*

Weightage : 70%

Note : *(i) Question No. **1** is compulsory.*

*(ii) Attempt any **three** questions from the rest.*

1. (a) Consider the following set of processes (P_1, P_2, P_3, P_4, P_5) with the length of the CPU burst time given in milliseconds along with the arrival time :

Process	Arrival Time	CPU Burst Time
P_1	1	6
P_2	2	4
P_3	4	1
P_4	3	2
P_5	5	3

Draw Gantt charts illustrating the execution of the processes using FCFS, SJF and RR (with quantum = 2) scheduling algorithms. Also find the average turn around time, average waiting time, processor utilization and throughput for all the algorithms. 10

(b) Following is the sequence of page requests :

1, 3, 2, 4, 3, 1, 4, 5, 2, 1, 3

Assume that there are three frames. How many page faults will occur if the following algorithms are used to replace the pages ? 10

(i) Least Recently Used (LRU)

(ii) First Come First Serve (FCFS)

(c) Define a semaphore. Using semaphores, write and explain the algorithm that solves the Reader's and Writer's problem. 10

(d) Discuss the security features in LINUX with respect to the following : 10

(i) User Accounts

(ii) File Permissions

(iii) Encrypted Storage

(iv) Remote Access

2. (a) There are three main alternative ways to structure a distributed application as shown below : 10

   (i) Client/Server (and RPC)

   (ii) Distributed objects

   (iii) Distributed shared memory

   Explain all the three models with the help of block diagrams respectively.

   (b) Discuss the following mechanisms/ techniques suitable for synchronization in multiprocessors : 10

   (i) Test-and-set

   (ii) Compare-and-swap

3. (a) Describe the evolution of iOS. Also, discuss the features of iOS operating system. 10

   (b) Explain the process management and process scheduling in Windows-10. 10

4. (a) With reference to wireless communication, explain the following 'two-unit configuration' ways of device communication : 10

   (i) Infrared connection

(ii) Bluetooth connection

(iii) Near Field Communication (NFC)

Also mention the drawbacks, if any, with the 'two-unit configuration' model/ technique.

(b) Explain the memory management in Android Operating System. 10

5. Write short notes on the following : 4×5=20

(a) Demand segmentation

(b) Deadlock avoidance

(c) Windows *vs.* Linux operating system

(d) Remote Procedure Call (RPC)